

Master Aptitude Preparation Guide

Formulas • Solved Examples • Practice Questions • Most-Asked Placement Questions

How to use this PDF

- Pages 1–6: Complete formula bank, topic-wise (revise this the night before any exam)
- Next: Every topic explained with 5 solved examples of five different question types
- Then: 5 practice questions per topic (answer key included at the end of each topic)
- Final section: Most frequently repeated aptitude questions across campus placement drives (TCS, Infosys, Wipro, Accenture, Capgemini, Cognizant, and general drives)

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Each topic follows the same structure: concept intro → 5 solved examples (5 different question types) → 5 practice questions with an answer key.

Complete Formula Bank — All Topics

Every formula you need, grouped by topic, with a one-line note on when to reach for it. Skim this section on exam morning.

1. Numbers, HCF & LCM

HCF × LCM = Product of two numbers

Use when given HCF, LCM and one number to find the other.

Sum of first n natural numbers = $n(n+1)/2$

Series and average-of-numbers questions.

Sum of squares = $n(n+1)(2n+1)/6$

Number-series based DI questions.

Sum of cubes = $[n(n+1)/2]^2$

Advanced number-series problems.

Divisibility: by 3/9 → digit sum divisible; by 4 → last 2 digits; by 8 → last 3 digits; by 11 → (sum of odd-place – sum of even-place digits) divisible by 11

Quick elimination in number-system MCQs.

Remainder theorem: $(a \times b) \bmod n = [(a \bmod n) \times (b \bmod n)] \bmod n$

Large-number remainder questions.

2. Percentages

Percentage = $(\text{Value} / \text{Total}) \times 100$

Base conversion for every percentage problem.

Increase/Decrease % = $(\text{Change} / \text{Original}) \times 100$

Population, price, marks-change questions.

If A is $x\%$ more than B, then B is $[x/(100+x)] \times 100\%$ less than A

Comparison-type percentage traps.

Successive % change = $x + y + xy/100$

Two back-to-back discounts/hikes.

Net effect of % increase then same % decrease = $-x^2/100$

"Price rises $x\%$ then falls $x\%$ " type shortcut.

3. Profit, Loss & Discount

Profit % = $(SP - CP)/CP \times 100$ Loss % = $(CP - SP)/CP \times 100$

Core profit/loss identity.

$SP = CP(1 + P\%/100)$ $SP = CP(1 - L\%/100)$

Finding SP directly from CP and %.

Marked Price & Discount: $SP = MP(1 - D\%/100)$

Discount-on-MP questions.

Successive discounts $d_1, d_2 \rightarrow$ effective = $d_1 + d_2 - (d_1 d_2 / 100)$

"20% then 10% off" combined discount.

False weight profit % = $(\text{True} - \text{False}) / \text{False} \times 100$

Shopkeeper-cheats-with-weight problems.

4. Simple & Compound Interest

$$SI = P \times R \times T / 100$$

Basic simple interest.

$$CI = P(1+R/100)^T - P$$

Basic compound interest, annual compounding.

$$\text{Amount (CI, n times/yr)} = P(1 + R/(100n))^{nT}$$

Half-yearly/quarterly compounding.

$$CI - SI \text{ (2 years)} = P(R/100)^2$$

Fast shortcut for 2-year difference questions.

$$CI - SI \text{ (3 years)} = P(R/100)^2[(300+R)/100]$$

3-year difference shortcut.

5. Ratio, Proportion & Partnership

$$a:b = a/b; \text{ if } a:b=c:d \text{ then } ad = bc$$

Cross-multiplication for proportion checks.

$$\text{Mean proportional} = \sqrt{ab}$$

"Insert a mean proportional between..." questions.

$$\text{Partnership profit share} \propto (\text{capital} \times \text{time})$$

Splitting profit among partners with unequal investment/duration.

$$\text{Compounded ratio of } a:b \text{ and } c:d = ac:bd$$

Chained ratio problems.

6. Averages, Mixtures & Alligation

$$\text{Average} = \text{Sum of items} / \text{Number of items}$$

Core average identity.

$$\text{New average when one value replaced: change in sum} / \text{count}$$

"One member leaves/joins" average shift.

$$\text{Alligation rule: (Cheaper qty)/(Dearer qty)} = (\text{Dearer price} - \text{Mean price})/(\text{Mean price} - \text{Cheaper price})$$

Mixing two ingredients/liquids to hit a target price or concentration.

$$\text{Replacement dilution: final conc.} = P(1 - x/V)^n$$

Repeated removal-and-replacement of milk/water.

7. Time & Work (incl. Pipes & Cisterns)

$$\text{Work rate} = 1/(\text{days to finish alone})$$

Convert "X finishes in n days" into a rate.

$$\text{Combined time (A+B)} = (A \times B)/(A+B) \text{ days}$$

Two people/pipes working together.

$$\text{Efficiency} \propto 1/\text{time}; \text{ work} = \text{men} \times \text{days} \times \text{hours}$$

Comparing efficiencies, "M1D1H1 = M2D2H2" chain rule.

$$\text{Pipe filling + pipe emptying together: } 1/T = 1/T_{\text{fill}} - 1/T_{\text{empty}}$$

Inlet/outlet pipe combined problems.

8. Time, Speed & Distance (incl. Trains, Boats)

Speed = Distance/Time; $\text{km/hr} \times 5/18 = \text{m/s}$

Core conversion, always check units first.

Average speed (equal distance, two speeds) = $2xy/(x+y)$

Trap question — never simple average.

Train crossing pole: L/S ; crossing platform: $(L_1+L_2)/S$

Train-length problems.

Relative speed: same direction $\rightarrow$ difference; opposite $\rightarrow$ sum of speeds

Two trains/two people crossing or catching up.

Boats: Downstream = $b+s$, Upstream = $b-s$; $b=(d+u)/2$, $s=(d-u)/2$

Boat & stream problems.

9. Ages

Set up present ages as x , kx (ratio form) then apply the "n years ago/hence" condition

Standard technique for every age problem.

10. Permutations, Combinations & Probability

$nPr = n!/(n-r)!$

Arrangements where order matters.

$nCr = n!/[r!(n-r)!]$

Selections where order doesn't matter.

Circular permutation of n objects = $(n-1)!$

Seating around a round table.

$P(E) = \text{Favourable outcomes} / \text{Total outcomes}$

Core probability definition.

$P(A \text{ or } B) = P(A)+P(B)-P(A \text{ and } B)$; independent: $P(A \text{ and } B)=P(A)P(B)$

Combined-event probability.

11. Algebra & Quadratic Equations

Roots of $ax^2+bx+c=0$: $x = [-b \pm \sqrt{(b^2-4ac)}]/2a$

Any quadratic equation question.

Sum of roots = $-b/a$; Product of roots = c/a

Fast root-relationship questions without solving fully.

$(a+b)^2=a^2+2ab+b^2$; $(a-b)^2=a^2-2ab+b^2$; $a^2-b^2=(a+b)(a-b)$

Simplification shortcuts in algebra MCQs.

12. Geometry & Mensuration

Triangle area = $\frac{1}{2} \times \text{base} \times \text{height}$; Heron's: $\sqrt{[s(s-a)(s-b)(s-c)]}$

Area from sides or base-height.

Circle: Area= πr^2 , Circumference= $2\pi r$

Circular field/track/wheel problems.

Rectangle/Square: Area= $l \times b$ / a^2 ; Perimeter= $2(l+b)$ / $4a$

Basic quadrilateral mensuration.

Cuboid: $V=lbh$; Cube: $V=a^3$; Cylinder: $V=\pi r^2h$; Sphere: $V=4/3\pi r^3$

3-D volume/surface-area questions.

Cone: $V=\frac{1}{3}\pi r^2h$, Curved $SA=\pi rl$, $l=\sqrt{r^2+h^2}$

Cone-shaped container problems.

13. Logarithms & Number Series

$\log(ab)=\log a+\log b$; $\log(a/b)=\log a-\log b$; $\log(a^n)=n \log a$

Log simplification identities.

$\log_b a = \log a / \log b$ (change of base)

Switching between log bases.

Series pattern-finding: check differences, ratios, alternating pattern, or squares/cubes offset

"Find the next number" and "find the odd one out" questions.

14. Clocks & Calendars

Angle between hands = $|30H - 11M/2|$ degrees

Angle-between-hands questions.

Hands coincide/opposite every $65\frac{5}{11}$ minutes

"How many times do hands overlap" questions.

Odd days: 1 ordinary yr=1, leap yr=2; century=5(100yr),3(200yr),1(300yr),0(400yr) odd days

Finding the day of the week for a given date.

Topic 1: Numbers, HCF & LCM

Covers divisibility rules, remainder-based questions, and HCF/LCM word problems — a near-guaranteed presence in every company aptitude round.

Solved Examples (5 different question types)

Example 1 — Basic HCF/LCM word problem

Find the greatest number that divides 245 and 1029, leaving remainder 5 in each case.

Solution: Subtract the remainder first: $245 - 5 = 240$, $1029 - 5 = 1024$. The required number is $\text{HCF}(240, 1024)$. $240 = 2^4 \times 3 \times 5$, $1024 = 2^{10}$. $\text{HCF} = 2^4 = 16$.

Answer: 16

Example 2 — LCM with "bells ring together" pattern

Three bells ring at intervals of 12, 15 and 18 minutes. If they ring together at 9:00 AM, when will they next ring together?

Solution: Find $\text{LCM}(12, 15, 18)$. $12 = 2^2 \times 3$, $15 = 3 \times 5$, $18 = 2 \times 3^2$. $\text{LCM} = 2^2 \times 3^2 \times 5 = 180$ minutes = 3 hours.

Answer: 12:00 PM

Example 3 — Divisibility rule application

Find the smallest digit to replace * so that $47*8$ is divisible by 3.

Solution: Digit sum = $4 + 7 + * + 8 = 19 + *$. Smallest * making $19 + *$ divisible by 3 is $* = 2$ (sum = 21).

Answer: 2

Example 4 — Remainder theorem on large exponents

Find the remainder when 7^{103} is divided by 5.

Solution: Powers of 7 mod 5 cycle as $7 \equiv 2$, $2^1 = 2$, $2^2 = 4$, $2^3 = 3$, $2^4 = 1$ (cycle length 4). $103 \bmod 4 = 3$, so remainder = $2^3 \bmod 5 = 3$.

Answer: 3

Example 5 — Numbers in ratio with given HCF

Two numbers are in the ratio 3:4 and their HCF is 5. Find their LCM.

Solution: Numbers = $3 \times 5 = 15$ and $4 \times 5 = 20$. $\text{LCM} = (15 \times 20) / \text{HCF} = 300 / 5 = 60$.

Answer: 60

Practice Questions

- Find the HCF and LCM of 84 and 126.
- Four bells toll at intervals of 6, 8, 12 and 18 seconds. In 1 hour, how many times will they toll together (excluding the start)?
- Find the remainder when 2^{50} is divided by 7.
- The least number which when divided by 6, 8 and 12 leaves remainder 3 in each case is?
- Two numbers are in ratio 5:6 and their LCM is 480. Find their sum.

Answers: 1) $\text{HCF} = 42$, $\text{LCM} = 252$ 2) 50 times 3) 4 4) 27 5) 176

Topic 2: Percentages

The foundation topic — almost every profit/loss, interest, and DI question is percentages in disguise.

Solved Examples (5 different question types)

Example 1 — Direct percentage of a number

What is 42% of 250?

Solution: $42/100 \times 250 = 105$.

Answer: 105

Example 2 — Percentage change (increase/decrease)

A number is first increased by 20% and then decreased by 20%. Find the net % change.

Solution: Net change = $x + y + xy/100$ with $x=20$, $y=-20$: $20-20+(20 \times -20)/100 = -4\%$.

Answer: 4% decrease

Example 3 — Comparison ("A is x% more than B")

If A's salary is 25% more than B's, by what % is B's salary less than A's?

Solution: B less than A by $[x/(100+x)] \times 100 = [25/125] \times 100 = 20\%$.

Answer: 20%

Example 4 — Reverse percentage (find the original)

After a 15% pay cut, an employee earns ₹25,500. What was the original salary?

Solution: Let original = x . $x(1-0.15)=25500 \Rightarrow 0.85x=25500 \Rightarrow x=30000$.

Answer: ₹30,000

Example 5 — Population/exam-pass style percentage problem

In an exam, 35% of students failed in Maths, 25% failed in English, and 10% failed in both. Find the % who passed in both.

Solution: Failed in at least one = $35+25-10=50\%$. Passed in both = $100-50=50\%$.

Answer: 50%

Practice Questions

1. What percentage of 640 is 96?
2. The price of an item is increased by 30% and then decreased by 30%. Find the net % change.
3. A's income is 40% less than B's. By what % is B's income more than A's?
4. A number decreased by 20% gives 96. Find the number.
5. In an election between two candidates, the winner got 60% of votes and won by 3,200 votes. Find the total votes polled.

Answers: 1) 15% 2) 9% decrease 3) 66.67% 4) 120 5) 16,000

Topic 3: Profit, Loss & Discount

Direct extension of percentages applied to buying/selling — frequent in TCS, Infosys and bank PO exams.

Solved Examples (5 different question types)

Example 1 — Basic profit percentage

A man buys an article for ₹450 and sells it for ₹540. Find his profit %.

Solution: Profit = 90. Profit% = $90/450 \times 100 = 20\%$.

Answer: 20%

Example 2 — Finding SP from CP and desired profit%

Find the SP of an article bought for ₹800 to gain 15%.

Solution: $SP = 800 \times 1.15 = 920$.

Answer: ₹920

Example 3 — Marked price and discount

A shopkeeper marks an item 40% above cost price and gives a 25% discount. Find his profit %.

Solution: Let CP=100. MP=140. $SP=140 \times 0.75=105$. Profit%=5%.

Answer: 5%

Example 4 — Successive discounts

Find the single equivalent discount for successive discounts of 20% and 10%.

Solution: Effective = $20+10-(20 \times 10)/100 = 28\%$.

Answer: 28%

Example 5 — False weight / cheating-type problem

A dishonest dealer professes to sell at cost price but uses a weight of 900g for 1kg. Find his profit %.

Solution: Profit% = $(\text{True}-\text{False})/\text{False} \times 100 = (1000-900)/900 \times 100 = 11.11\%$.

Answer: 11.11%

Practice Questions

1. An article is sold at a loss of 12% for ₹528. Find its cost price.
2. By selling 45 lemons for ₹40, a person loses 20%. How many should he sell for ₹24 to gain 20%?
3. A trader marks his goods 50% above cost and offers a discount of 20%. Find his profit %.
4. Find the single discount equivalent to two successive discounts of 10% and 15%.
5. A vendor buys 12 oranges for ₹10 and sells 10 for ₹12. Find his profit %.

Answers: 1) ₹600 2) 18 lemons 3) 20% 4) 23.5% 5) 44%

Topic 4: Simple & Compound Interest

Bank/finance-flavoured but common across all recruiters. Watch for the CI–SI shortcut — it saves the most time.

Solved Examples (5 different question types)

Example 1 — Basic simple interest

Find the SI on ₹12,000 at 9% p.a. for 3 years.

Solution: $SI = 12000 \times 9 \times 3 / 100 = ₹3,240$.

Answer: ₹3,240

Example 2 — Basic compound interest

Find the CI on ₹10,000 at 10% p.a. for 2 years, compounded annually.

Solution: Amount = $10000 \times 1.1^2 = 12,100$. CI = $12100 - 10000 = ₹2,100$.

Answer: ₹2,100

Example 3 — CI–SI difference (2-year shortcut)

The difference between CI and SI on a sum for 2 years at 10% p.a. is ₹150. Find the sum.

Solution: Difference = $P(R/100)^2 \Rightarrow 150 = P(0.01) \Rightarrow P = ₹15,000$.

Answer: ₹15,000

Example 4 — Half-yearly compounding

Find the CI on ₹8,000 at 10% p.a. for 1 year compounded half-yearly.

Solution: Rate/period = 5%, periods = 2. Amount = $8000 \times 1.05^2 = 8,820$. CI = ₹820.

Answer: ₹820

Example 5 — Finding rate or time from given SI

At what rate % p.a. will ₹5,000 amount to ₹6,050 as SI in 3 years?

Solution: SI = 1050. $R = SI \times 100 / (P \times T) = 1050 \times 100 / (5000 \times 3) = 7\%$.

Answer: 7%

Practice Questions

- Find the SI on ₹7,500 at 12% p.a. for 4 years.
- Find the CI on ₹20,000 at 5% p.a. for 3 years, compounded annually.
- The difference between CI and SI on a sum for 3 years at 5% p.a. is ₹76.25. Find the sum.
- A sum doubles itself in 8 years at simple interest. Find the rate %.
- Find the CI on ₹15,000 at 8% p.a. for 2 years compounded annually.

Answers: 1) ₹3,600 2) ₹3,152.50 3) ₹10,000 4) 12.5% 5) ₹2,496

Topic 5: Ratio, Proportion & Partnership

Underlies age, mixture and profit-sharing problems — master the "parts" technique here.

Solved Examples (5 different question types)

Example 1 — Dividing a quantity in a given ratio

Divide ₹2,400 among A, B, C in the ratio 2:3:5.

Solution: Total parts = 10. $A = 2400 \times 2/10 = 480$, $B = 720$, $C = 1200$.

Answer: A=₹480, B=₹720, C=₹1,200

Example 2 — Ratio with an added/removed quantity

Two numbers are in ratio 3:5. If 10 is added to both, the ratio becomes 5:7. Find the numbers.

Solution: Let numbers = $3x, 5x$. $(3x+10)/(5x+10) = 5/7 \Rightarrow 21x+70=25x+50 \Rightarrow 4x=20 \Rightarrow x=5$. Numbers = 15, 25.

Answer: 15 and 25

Example 3 — Partnership with equal time, unequal capital

A and B invest ₹30,000 and ₹45,000 in a business. If total profit is ₹35,000, find B's share.

Solution: Ratio = $30000:45000 = 2:3$. B's share = $35000 \times 3/5 = ₹21,000$.

Answer: ₹21,000

Example 4 — Partnership with unequal capital and time

A invests ₹20,000 for 12 months, B invests ₹15,000 for 8 months. Divide a profit of ₹9,600.

Solution: Ratio = $(20000 \times 12):(15000 \times 8) = 240000:120000 = 2:1$. A gets ₹6,400, B gets ₹3,200.

Answer: A=₹6,400, B=₹3,200

Example 5 — Mean/third/fourth proportional

Find the fourth proportional to 4, 9, 12.

Solution: $4:9 = 12:x \Rightarrow x = 9 \times 12/4 = 27$.

Answer: 27

Practice Questions

- Divide ₹1,800 among A, B, C so that $A:B=2:3$ and $B:C=4:5$.
- The ratio of two numbers is 4:7. If each number is increased by 6, the ratio becomes 5:8. Find the numbers.
- A, B and C invest in a ratio of 3:5:7 for the same period. If the total profit is ₹33,000, find C's share.
- A starts a business with ₹40,000. After 4 months, B joins with ₹60,000. Find the ratio of their profit shares at year end.
- Find the mean proportional between 16 and 25.

Answers: 1) $A=₹480, B=₹720, C=₹900$ 2) 24 and 42 3) ₹15,400 4) 1:1 5) 20

Topic 6: Averages, Mixtures & Alligation

Alligation is the single highest-leverage shortcut in this section — it collapses long mixture equations into one line.

Solved Examples (5 different question types)

Example 1 — Basic average

Find the average of 12, 18, 24, 30, 36.

Solution: Sum=120, count=5. Average=24.

Answer: 24

Example 2 — Average with replacement of one member

The average age of 30 students is 12 years. If the teacher's age (40) is included, the average becomes 13. Verify / find new average check.

Solution: Old sum=360. New sum=360+40=400 over 31 students. New average=400/31 $\approx$ 12.9 $\approx$ 13.

Answer: $\approx$ 12.9 years

Example 3 — Alligation for mixing two priced items

In what ratio must tea worth ₹60/kg be mixed with tea worth ₹80/kg so the mixture costs ₹65/kg?

Solution: Alligation: (80–65):(65–60) = 15:5 = 3:1.

Answer: 3:1 (cheaper:dearer)

Example 4 — Replacement/dilution problem

A container has 40L milk. 4L is removed and replaced with water, and this is repeated once more. Find the final quantity of milk.

Solution: Final milk = $40(1-4/40)^2 = 40 \times 0.9^2 = 32.4\text{L}$.

Answer: 32.4 L

Example 5 — Weighted average of groups

Class A (30 students) averages 70 marks, Class B (20 students) averages 80 marks. Find the combined average.

Solution: Combined avg = $(30 \times 70 + 20 \times 80) / 50 = (2100 + 1600) / 50 = 74$.

Answer: 74

Practice Questions

- Find the average of the first 20 natural numbers.
- The average weight of 8 people increases by 2.5kg when a new person replaces one weighing 65kg. Find the new person's weight.
- In what ratio should rice at ₹40/kg be mixed with rice at ₹60/kg to get a mixture worth ₹52/kg?
- A vessel contains 60L of milk. 12L is withdrawn and replaced with water, repeated once more. Find the milk remaining.
- The average of 5 consecutive odd numbers is 41. Find the largest number.

Answers: 1) 10.5 2) 85 kg 3) 2:3 4) 38.4 L 5) 45

Topic 7: Time & Work (incl. Pipes & Cisterns)

Always convert to "work per day" rate first — every variant (men leaving, pipes, alternate days) becomes simple addition after that.

Solved Examples (5 different question types)

Example 1 — Two people working together

A can do a work in 12 days, B in 18 days. How long will they take working together?

Solution: Combined rate = $1/12 + 1/18 = 5/36$. Time = $36/5 = 7.2$ days.

Answer: 7.2 days

Example 2 — Efficiency comparison

A is twice as efficient as B. Together they finish a job in 12 days. Find the time A alone takes.

Solution: Let B's rate = x , A's rate = $2x$. $3x = 1/12 \Rightarrow x = 1/36$. A's time = $1/2x = 18$ days.

Answer: 18 days

Example 3 — Work with a person leaving midway

A and B together can finish a job in 10 days. They work together for 4 days, then A leaves; B finishes the remaining work in 9 days. Find B's time alone.

Solution: Work done together in 4 days = $4/10 = 2/5$. Remaining = $3/5$ done by B in 9 days $\Rightarrow$ B's rate = $(3/5)/9 = 1/15$. B alone = 15 days.

Answer: 15 days

Example 4 — Pipes: inlet + outlet together

Pipe A fills a tank in 6 hrs, pipe B empties it in 9 hrs. If both are opened together, how long to fill the tank?

Solution: Net rate = $1/6 - 1/9 = 1/18$. Time = 18 hours.

Answer: 18 hours

Example 5 — Men-days-hours chain rule

12 men working 8 hrs/day finish a job in 10 days. How many days will 16 men working 6 hrs/day take for the same job?

Solution: $M_1D_1H_1 = M_2D_2H_2 \Rightarrow 12 \times 10 \times 8 = 16 \times D_2 \times 6 \Rightarrow 960 = 96D_2 \Rightarrow D_2 = 10$ days.

Answer: 10 days

Practice Questions

1. A can finish a work in 15 days and B in 10 days. How long together?
2. A is thrice as efficient as B, and together they finish a job in 15 days. Find B's time alone.
3. A, B and C together finish a job in 6 days. A alone takes 18 days, B alone takes 24 days. Find C's time alone.
4. Two pipes fill a tank in 20 and 30 minutes respectively; a third pipe empties it in 15 minutes. If all three are opened, find the time to fill the tank.
5. 15 workers can build a wall in 24 days. How many workers are needed to finish it in 18 days?

Answers: 1) 6 days 2) 60 days 3) 14.4 days 4) 60 minutes 5) 20 workers

Topic 8: Time, Speed & Distance (incl. Trains, Boats)

Unit conversion mistakes cause most errors here — always fix units (km/hr vs m/s) before plugging into a formula.

Solved Examples (5 different question types)

Example 1 — Basic speed-distance-time

A car covers 210 km in 3.5 hours. Find its speed.

Solution: Speed = $210/3.5 = 60$ km/hr.

Answer: 60 km/hr

Example 2 — Average speed trap (equal distances, two speeds)

A person travels a certain distance at 40 km/hr and returns at 60 km/hr. Find the average speed for the whole journey.

Solution: Average speed = $2 \times 40 \times 60 / (40 + 60) = 4800/100 = 48$ km/hr (NOT the plain average of 50).

Answer: 48 km/hr

Example 3 — Train crossing a platform

A train 150m long crosses a platform 250m long in 20 seconds. Find its speed in km/hr.

Solution: Total distance=400m in 20s $\Rightarrow$ speed=20 m/s = $20 \times 18/5 = 72$ km/hr.

Answer: 72 km/hr

Example 4 — Relative speed, two trains crossing

Two trains of length 120m and 180m run in opposite directions at 54 km/hr and 36 km/hr. Find the time to cross each other.

Solution: Relative speed = $(54+36) \times 5/18 = 25$ m/s. Total length=300m. Time=300/25=12 seconds.

Answer: 12 seconds

Example 5 — Boats and streams

A boat's speed downstream is 20 km/hr and upstream is 12 km/hr. Find the speed of the boat in still water and the stream's speed.

Solution: Boat speed = $(20+12)/2 = 16$ km/hr. Stream speed = $(20-12)/2 = 4$ km/hr.

Answer: Boat = 16 km/hr, Stream = 4 km/hr

Practice Questions

1. A cyclist covers 96 km in 6 hours. Find his speed in m/s.
2. A man travels one-third of a journey at 20 km/hr, the next third at 30 km/hr, and the last third at 40 km/hr. Find his average speed for the whole journey.
3. A train 200m long crosses a pole in 10 seconds. Find its speed in km/hr.
4. Two trains 110m and 90m long run on parallel tracks in the same direction at 60 km/hr and 42 km/hr. Find the time taken by the faster train to cross the slower one.
5. A boat covers 24 km upstream in 6 hours and the same distance downstream in 4 hours. Find the speed of the stream.

Answers: 1) 4.44 m/s 2) 27.7 km/hr (approx, exact = $360/13$) 3) 72 km/hr 4) 40 seconds 5) 1 km/hr

Topic 9: Problems on Ages

Always express ages as a single variable using the given ratio, then translate the "years ago/hence" sentence directly into an equation.

Solved Examples (5 different question types)

Example 1 — Present ages from a ratio and sum

The ages of A and B are in ratio 5:7. After 4 years, their sum of ages will be 56. Find their present ages.

Solution: Present ages = $5x, 7x$. $(5x+4)+(7x+4)=56 \Rightarrow 12x=48 \Rightarrow x=4$. Ages = 20, 28.

Answer: 20 and 28 years

Example 2 — "n years ago" condition

Father is 3 times as old as his son. 5 years ago, he was 4 times as old. Find their present ages.

Solution: Son= x , Father= $3x$. $3x-5=4(x-5) \Rightarrow 3x-5=4x-20 \Rightarrow x=15$. Son=15, Father=45.

Answer: Son=15, Father=45

Example 3 — Future age condition

A mother's age is 4 times her daughter's age. After 8 years, the mother will be twice as old. Find their present ages.

Solution: Daughter= x , Mother= $4x$. $4x+8=2(x+8) \Rightarrow 4x+8=2x+16 \Rightarrow x=4$. Daughter=4, Mother=16.

Answer: Daughter=4, Mother=16

Example 4 — Sum and difference of ages

The sum of ages of a father and son is 60. Six years ago father's age was 5 times son's age. Find their present ages.

Solution: Father= F , Son= S . $F+S=60$. $(F-6)=5(S-6) \Rightarrow F=5S-24$. Sub: $5S-24+S=60 \Rightarrow 6S=84 \Rightarrow S=14$, $F=46$.

Answer: Father=46, Son=14

Example 5 — Multi-generation age problem

A is 20 years older than B. In 10 years, A will be twice as old as B. Find B's present age.

Solution: $A=B+20$. $B+20+10=2(B+10) \Rightarrow B+30=2B+20 \Rightarrow B=10$.

Answer: 10 years

Practice Questions

1. The ages of P and Q are in ratio 3:4. Ten years hence, the ratio becomes 4:5. Find their present ages.
2. A father is 5 times as old as his son. After 4 years, he will be 4 times as old. Find their present ages.
3. The sum of present ages of a man and his wife is 100. Five years ago the ratio of their ages was 5:4. Find the man's present age.
4. A boy's age 4 years ago was one-third of his age 8 years from now. Find his present age.
5. Twenty years ago, a father was 3 times as old as his son. Now he is only 2 times as old. Find the father's present age.

Answers: 1) 30 and 40 2) Son=12, Father=60 3) 55 years 4) 10 years 5) 80 years

Topic 10: Permutations, Combinations & Probability

The trick is always deciding "does order matter?" first — that single decision routes you to nPr or nCr .

Solved Examples (5 different question types)

Example 1 — Basic arrangement (permutation)

In how many ways can the letters of the word "TABLE" be arranged?

Solution: 5 distinct letters $\Rightarrow 5! = 120$ ways.

Answer: 120

Example 2 — Selection (combination)

In how many ways can a committee of 3 be chosen from 8 people?

Solution: ${}^8C_3 = \frac{8!}{(3!5!)} = 56$.

Answer: 56

Example 3 — Arrangement with repeated letters / restriction

How many distinct arrangements are there of the letters of "STATISTICS"?

Solution: 10 letters with S=3, T=3, I=2, A=1, C=1. Arrangements = $\frac{10!}{(3!3!2!)} = 50,400$.

Answer: 50,400

Example 4 — Combined selection (men and women)

From 6 men and 4 women, in how many ways can a team of 4 be selected with exactly 2 women?

Solution: Choose 2 women from 4 (${}^4C_2=6$) and 2 men from 6 (${}^6C_2=15$). Total = $6 \times 15 = 90$.

Answer: 90

Example 5 — Probability of a compound event

Two dice are thrown. Find the probability that the sum is 8.

Solution: Favourable pairs: (2,6)(3,5)(4,4)(5,3)(6,2) = 5. Total outcomes = 36. $P = \frac{5}{36}$.

Answer: 5/36

Practice Questions

1. How many 4-letter words (with repetition allowed) can be formed from the 26 English letters?
2. In how many ways can 5 people be seated around a circular table?
3. From a deck of 52 cards, one card is drawn. Find the probability it is a king or a heart.
4. A bag has 5 red and 4 blue balls. Find the probability of drawing 2 red balls without replacement.
5. How many ways can a 3-digit number be formed using digits 1–9 without repetition?

Answers: 1) 456,976 2) 24 3) $\frac{4}{13}$ 4) $\frac{5}{18}$ 5) 504

Topic 11: Algebra & Quadratic Equations

Sum/product-of-roots shortcuts let you answer many MCQs without ever solving the full quadratic.

Solved Examples (5 different question types)

Example 1 — Solving a quadratic by factoring

Solve: $x^2 - 7x + 12 = 0$.

Solution: Factors of 12 summing to 7: 3 and 4. $(x-3)(x-4)=0 \Rightarrow x=3$ or 4.

Answer: $x = 3, 4$

Example 2 — Sum and product of roots (no solving needed)

If the roots of $x^2 - 5x + k = 0$ are equal, find k .

Solution: Equal roots $\Rightarrow$ discriminant $= 0$: $25 - 4k = 0 \Rightarrow k = 6.25$.

Answer: $k = 6.25$

Example 3 — Word problem forming a linear equation

The sum of two numbers is 45 and their difference is 9. Find the numbers.

Solution: $x+y=45$, $x-y=9$. Adding: $2x=54 \Rightarrow x=27$, $y=18$.

Answer: 27 and 18

Example 4 — Algebraic identity simplification

If $x + 1/x = 5$, find the value of $x^2 + 1/x^2$.

Solution: $x^2 + 1/x^2 = (x + 1/x)^2 - 2 = 25 - 2 = 23$.

Answer: 23

Example 5 — Simultaneous (linear) equations word problem

5 chairs and 3 tables cost ₹7,000. 3 chairs and 5 tables cost ₹9,000. Find the cost of one table.

Solution: $5c+3t=7000$, $3c+5t=9000$. Multiply first by 5, second by 3: $25c+15t=35000$; $9c+15t=27000$. Subtract: $16c=8000 \Rightarrow c=500$. Then $t=(7000-2500)/3=1500$.

Answer: ₹1,500

Practice Questions

1. Solve: $x^2 - 9x + 20 = 0$.
2. If one root of $x^2 - 6x + k = 0$ is 2, find the other root and k .
3. The sum of three consecutive integers is 72. Find them.
4. If $x - 1/x = 3$, find $x^2 + 1/x^2$.
5. Twice a number added to 3 times another equals 22; the numbers differ by 1. Find both numbers.

Answers: 1) $x=4, 5$ 2) other root $= 4$, $k=8$ 3) 23, 24, 25 4) 11 5) 5 and 4

Topic 12: Geometry & Mensuration

Draw a quick sketch for every geometry question — it prevents almost all careless formula-selection errors.

Solved Examples (5 different question types)

Example 1 — Triangle area from sides (Heron's formula)

Find the area of a triangle with sides 13, 14, 15.

Solution: $s = (13 + 14 + 15)/2 = 21$. Area $= \sqrt{21(21-13)(21-14)(21-15)} = \sqrt{21 \times 8 \times 7 \times 6} = \sqrt{7056} = 84$.

Answer: 84 sq. units

Example 2 — Circle: area and circumference

A circular garden has a radius of 14m. Find its area and circumference. ($\pi = 22/7$)

Solution: Area $= \pi r^2 = 22/7 \times 196 = 616 \text{ m}^2$. Circumference $= 2\pi r = 88\text{m}$.

Answer: Area=616 m², Circumference=88 m

Example 3 — Rectangle/square perimeter-area relation

The length of a rectangle is twice its breadth. If its perimeter is 60cm, find its area.

Solution: $l = 2b$. $2(2b + b) = 60 \Rightarrow 6b = 60 \Rightarrow b = 10, l = 20$. Area $= 200 \text{ cm}^2$.

Answer: 200 cm²

Example 4 — 3-D volume problem

Find the volume and total surface area of a cube with side 6cm.

Solution: Volume $= 6^3 = 216 \text{ cm}^3$. TSA $= 6 \times 6^2 = 216 \text{ cm}^2$.

Answer: Volume=216 cm³, TSA=216 cm²

Example 5 — Cylinder capacity problem

A cylindrical tank has radius 7m and height 10m. Find its volume. ($\pi = 22/7$)

Solution: Volume $= \pi r^2 h = 22/7 \times 49 \times 10 = 1,540 \text{ m}^3$.

Answer: 1,540 m³

Practice Questions

- Find the area of a triangle with sides 9, 40, 41.
- A circular wheel of radius 0.7m makes 1000 revolutions. Find the distance covered ($\pi = 22/7$).
- The perimeter of a square is 64cm. Find its area.
- Find the curved surface area and volume of a cone with radius 7cm and height 24cm.
- A rectangular water tank is 6m $\times$ 5m $\times$ 4m. Find its capacity in litres.

Answers: 1) 180 sq. units 2) 4,400 m 3) 256 cm² 4) CSA=550 cm², Volume=1,232 cm³ 5) 1,20,000 litres

Topic 13: Logarithms, Number Series & Data Interpretation Basics

Series questions reward pattern-spotting speed; DI questions reward careful reading of the table/graph before calculating.

Solved Examples (5 different question types)

Example 1 — Basic logarithm evaluation

Find the value of $\log_2 32$.

Solution: $32=2^5$, so $\log_2 32=5$.

Answer: 5

Example 2 — Log identity simplification

Simplify: $\log 45 - \log 9 + \log 4$.

Solution: $= \log(45 \times 4/9) = \log(180/9) = \log 20$.

Answer: $\log 20$

Example 3 — Number series (find next term)

Find the next term: 2, 6, 12, 20, 30, ?

Solution: Differences are 4,6,8,10 (increasing by 2). Next difference=12. Next term= $30+12=42$. (Pattern is $n(n+1)$.)

Answer: 42

Example 4 — Odd one out in a series

Find the odd one out: 3, 5, 11, 14, 17, 21.

Solution: All others are odd numbers except 14, which is even. 14 is the odd one out (breaks the "all odd numbers" pattern).

Answer: 14

Example 5 — Simple data interpretation (table reading)

A company's sales (in ₹lakh) were: Jan-40, Feb-55, Mar-45, Apr-60. Find the % increase from Jan to Apr.

Solution: Increase= $60-40=20$. %increase= $20/40 \times 100=50\%$.

Answer: 50%

Practice Questions

- Find the value of $\log_5 125$.
- If $\log 2=0.301$, find $\log 8$.
- Find the next term: 1, 4, 9, 16, 25, ?
- Find the odd one out: 4, 9, 16, 26, 36, 49.
- A student's marks in 5 subjects are 60, 72, 55, 80, 68 (out of 100 each). Find the overall percentage.

Answers: 1) 3 2) 0.903 3) 36 4) 26 5) 67%

Topic 14: Clocks & Calendars

Low-frequency but high-scoring when they do appear — the angle formula and odd-days method cover almost every variant.

Solved Examples (5 different question types)

Example 1 — Angle between hour and minute hands

Find the angle between the hands of a clock at 4:20.

Solution: $\text{Angle} = |30H - 11M/2| = |30 \times 4 - 11 \times 20/2| = |120 - 110| = 10^\circ$.

Answer: 10°

Example 2 — Time when hands are opposite (180°)

At what time between 7 and 8 o'clock will the hands of a clock be exactly opposite each other?

Solution: At 7:00 hour hand is at 210° . Need minute hand 180° away. Using formula, opposite occurs at $7:05\frac{5}{11}$.

Answer: $7:05\frac{5}{11}$

Example 3 — Clock gain/loss problem

A clock gains 5 minutes every hour. If it is set right at 12:00 noon, what will it show at 12:00 midnight (true time)?

Solution: In 12 true hours, clock gains $5 \times 12 = 60$ minutes = 1 hour. Clock shows 1:00 AM.

Answer: 1:00 AM

Example 4 — Day of the week (odd days method)

If 1st January 2024 was a Monday, what day was 1st January 2025?

Solution: 2024 is a leap year $\Rightarrow$ 2 odd days. $\text{Monday} + 2 = \text{Wednesday}$.

Answer: Wednesday

Example 5 — Calendar repeat-year problem

What is the least number of years after which the calendar of a normal (non-leap) year repeats itself?

Solution: Calendar of a non-leap year repeats after 6 years if no leap year occurs in between; with the usual leap-year pattern the calendar repeats after 6 or 11 years depending on the position of leap years — the standard exam answer is 6 years (when no leap year falls in the gap) or 11/28 in general cases.

Answer: 6 years (standard case)

Practice Questions

- Find the angle between the hands of a clock at 3:15.
- How many times do the hands of a clock coincide in a day (24 hours)?
- A watch loses 3 minutes every hour. If set right at 8:00 AM, what will it show at 8:00 PM (true time) the same day?
- If 1st March 2023 was a Wednesday, what day was 1st March 2024?
- At what time between 3 and 4 o'clock will the hour and minute hands be at right angles for the first time?

Answers: 1) 7.5° 2) 22 times 3) 7:24 PM 4) Friday 5) $3:32\frac{8}{11}$

Final Section: Most-Asked Placement Aptitude Questions

These are the question patterns that repeat most often across TCS NQT, Infosys, Wipro, Accenture, Capgemini and Cognizant aptitude rounds, plus common bank-PO style questions. Skim this section right before your exam.

Numbers & Percentages

TCS / INFOSYS PATTERN

A number when divided by 899 gives a remainder 63. What remainder will be obtained by dividing the same number by 29?

Answer: 5 (since $899=29\times31$, remainder $63 \bmod 29 = 5$)

WIPRO / ACCENTURE PATTERN

If the price of sugar rises by 25%, by what % must a family reduce consumption to keep expenditure unchanged?

Answer: 20% (formula: $x/(100+x)\times100$)

CAPGEMINI PATTERN

Two numbers are respectively 20% and 50% more than a third number. Find the ratio of the two numbers.

Answer: 4:5

Profit, Loss, SI/CI

TCS PATTERN

A shopkeeper sells an item at 10% loss. If he had sold it for ₹60 more, he would have gained 5%. Find the cost price.

Answer: ₹400

COGNIZANT PATTERN

A sum becomes ₹7,350 in 2 years and ₹8,575 in 3 years at compound interest. Find the rate of interest.

Answer: 16.67% (approx)

Time, Speed, Distance & Work

INFOSYS PATTERN

A train travels at 60 km/hr without stoppage and at 50 km/hr with stoppage. Find the stoppage time per hour.

Answer: 10 minutes

ACCENTURE PATTERN

A and B can do a job in 20 and 30 days respectively. They start together, but A leaves 5 days before completion. Find total days taken.

Answer: 15 days

WIPRO PATTERN

Two trains start simultaneously from stations 300km apart towards each other at 40 km/hr and 60 km/hr. When do they meet?

Answer: 3 hours

Ratio, Averages & Ages

TCS NQT PATTERN

The average of 11 results is 50. If the average of the first 6 is 49 and the last 6 is 52, find the 6th result.

Answer: 56

CAPGEMINI PATTERN

Ten years ago, the ages of a father and son were in ratio 3:1. Ten years hence, the ratio will be 2:1. Find their present ages.

Answer: Father=70, Son=30

Permutations, Probability & Series

INFOSYS / COGNIZANT PATTERN

A box has 4 red, 5 green and 6 blue balls. Find the probability of drawing 1 ball of each colour when 3 balls are drawn.

Answer: 24/91

TCS PATTERN

Find the missing number in the series: 5, 11, 19, 29, 41, ?

Answer: 55 (differences: 6,8,10,12,14)

WIPRO PATTERN

In how many ways can 4 boys and 3 girls be seated in a row so that no two girls sit together?

Answer: $4! \times 5P3 = 24 \times 60 = 1,440$

Data Interpretation & Logical-Numeric Reasoning

ACCENTURE / CAPGEMINI PATTERN

In a class, the ratio of boys to girls is 5:3. If 10 more girls join, the ratio becomes 5:4. Find the number of boys.

Answer: 50

COGNIZANT PATTERN

A pie chart shows a company's expenses: Salaries 40%, Rent 15%, Marketing 20%, Others 25%. If total expense is ₹50 lakh, find the amount spent on Marketing.

Answer: ₹10 lakh

Classic "always comes back" questions

GENERAL — EVERY COMPANY

A man walks a certain distance and rides back in 5 hours 30 minutes. He could walk both ways in 7 hours 30 minutes. How long would it take him to ride both ways?

Answer: 3 hours 30 minutes

GENERAL — EVERY COMPANY

The sum of ages of 5 children born at 3-year intervals is 50. Find the age of the youngest child.

Answer: 4 years

GENERAL — EVERY COMPANY

A can complete a work in 10 days, B in 15 days. They work on alternate days starting with A. In how many days will the work be completed?

Answer: 12 days

GENERAL — EVERY COMPANY

Find the compound interest on ₹25,000 for 2 years at 8% p.a., compounded annually.

Answer: ₹4,160

GENERAL — EVERY COMPANY

A sum of money doubles itself in 5 years at simple interest. In how many years will it become 4 times itself?

Answer: 15 years

Exam-day checklist: revise the formula bank (pages 2-7) → skim one solved example per topic → re-attempt any practice question you got wrong earlier → finish with this placement-questions section. Good luck!